

MILWAUKEE MITCHELL, WI, USA

WMO: 726400

Lat: 42.955N

Lon: 87.904W

Elev: 204

StdP: 98.90

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.2	-25.3	0.4	-17.9	-22.7	0.5	-15.6	11.9	-3.7	10.8	-3.1	5.2	290	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	32.0	23.5	30.2	22.4	28.6	21.5	24.8	30.1	23.8	28.5	22.8	27.1	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	18.3	27.9	22.1	17.2	26.8	21.2	16.3	25.7	76.2	30.1	71.9	28.5	68.1	27.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.4	8.6	-22.3	34.8	3.8	2.1	-25.0	36.3	-27.2	37.5	-29.3	38.6	-32.1	40.1		
			-22.8	26.5	3.6	1.3	-25.4	27.4	-27.5	28.1	-29.6	28.9	-32.2	29.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-5.0	-3.2	2.0	7.6	13.5	19.4	22.5	22.0	18.1	11.5	4.3	-1.8	(6)
	DBStd	10.78	6.33	5.97	5.77	4.77	4.89	4.48	3.42	2.96	4.16	4.68	5.20	5.72	(7)
	HDD10.0	1801	464	371	257	102	18	0	0	0	1	38	182	366	(8)
	HDD18.3	3693	723	604	506	324	167	41	5	5	54	218	421	624	(9)
	CDD10.0	1547	0	1	10	31	126	282	388	371	244	84	11	1	(10)
	CDD18.3	397	0	0	1	2	16	73	135	118	48	6	0	0	(11)
	CDH23.3	2906	0	0	3	31	151	600	1081	709	296	36	0	0	(12)
	CDH26.7	860	0	0	0	6	35	194	358	188	73	6	0	0	(13)
(14) Wind	WSAvg	4.3	4.8	4.7	4.7	5.0	4.4	3.8	3.8	3.6	3.9	4.4	4.6	4.5	(14)
(15) Precipitation	PrecAvg	836	45	38	62	92	78	98	86	88	80	61	57	51	(15)
	PrecMax	1146	111	89	176	188	214	312	278	230	239	179	137	138	(16)
	PrecMin	485	8	1	8	33	13	18	18	14	0	4	9	7	(17)
	PrecStd	148	26	24	35	37	43	59	50	50	53	34	28	30	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	14.4	22.7	27.4	30.1	33.7	35.0	32.9	31.4	27.3	19.8	14.4	(19)
		MCWB	8.8	10.2	15.4	16.7	20.5	24.0	25.0	24.4	22.4	19.7	14.3	12.4	(20)
	2%	DB	7.4	9.8	17.4	22.7	27.3	31.2	32.2	30.6	28.7	23.7	16.6	10.3	(21)
		MCWB	5.0	6.3	12.5	14.9	19.0	22.4	24.2	23.0	21.2	17.7	12.2	7.9	(22)
	5%	DB	4.6	6.4	13.5	18.9	24.7	28.9	30.3	28.7	26.7	21.3	14.4	7.6	(23)
		MCWB	3.0	3.7	9.2	12.4	17.8	21.4	22.9	21.9	20.2	16.2	10.9	5.7	(24)
	10%	DB	3.1	4.1	10.0	15.7	22.1	26.8	28.5	27.1	24.6	18.7	11.8	5.5	(25)
		MCWB	1.9	2.2	6.5	10.3	16.3	20.3	21.8	21.2	19.5	14.7	8.7	3.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.7	16.7	18.5	22.2	25.3	26.8	25.5	23.9	21.4	15.3	12.5	(27)
		MCDB	10.6	13.1	21.9	25.2	28.0	31.5	32.4	30.9	29.2	25.8	18.1	14.2	(28)
	2%	WB	5.2	6.7	13.4	15.6	20.3	23.7	25.2	24.3	22.4	19.1	13.4	8.6	(29)
		MCDB	7.0	9.4	16.9	21.7	25.4	29.5	30.6	28.6	26.7	22.2	16.0	9.7	(30)
	5%	WB	3.2	4.1	9.7	13.3	18.8	22.3	24.0	23.2	21.5	17.1	11.0	5.9	(31)
		MCDB	4.5	6.2	13.0	18.4	23.5	27.7	28.8	27.0	25.2	20.2	13.6	7.3	(32)
	10%	WB	1.7	2.4	6.6	10.6	17.0	21.1	22.9	22.3	20.3	15.3	9.0	3.7	(33)
		MCDB	3.1	4.0	9.7	14.8	21.5	25.4	27.2	25.7	23.8	18.5	11.5	5.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.1	7.7	8.6	9.3	9.2	8.5	7.8	8.6	8.2	7.3	6.5	(35)
		MCDBR	8.4	9.7	12.6	15.0	13.2	11.6	10.4	9.9	10.9	11.7	10.6	8.7	(36)
	5% WB	MCWBR	7.3	7.3	9.0	9.1	7.7	6.1	5.1	4.8	5.4	7.1	7.8	7.3	(37)
		MCDWR	7.8	9.2	12.0	14.3	12.4	10.8	9.6	8.6	9.5	10.4	9.7	7.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.290	0.313	0.343	0.391	0.414	0.434	0.429	0.421	0.392	0.348	0.310	0.288	(40)
		taud	2.429	2.364	2.384	2.285	2.271	2.264	2.279	2.298	2.357	2.475	2.519	2.475	(41)
	Ebn at Noon		852	885	899	875	861	842	842	836	835	836	825	822	(42)
		Edh at Noon	75	95	106	127	132	133	130	124	108	85	69	66	(43)
(44) All-Sky Solar Radiation	RadAvg		1.49	2.41	3.58	4.58	5.32	6.03	6.14	5.29	4.28	2.75	1.72	1.21	(44)
	RadStd		0.20	0.28	0.33	0.40	0.42	0.41	0.28	0.33	0.36	0.31	0.16	0.13	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page